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Flight Program:			
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Key Personnel Changes/Previous PI:	Kristy Sauers, who was listed on the original team as NASA consultant with Johnson Space Center's Flight Deck of the Future, was replaced with Brian Schwing, also a NASA civil servant, due to a NASA organizational reassignment. Brian Schwing served in the same capacity as Kristy Sauers.		
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Task Description:

NASA has demonstrated an interest in improving astronaut health and performance through the installment of a new lighting countermeasure on the International Space Station. The Solid State Lighting Assembly (SSLA) system is designed to positively influence astronaut health by providing a daily change to light spectrum to improve circadian entrainment. Unfortunately, existing NASA standards and requirements define ambient light level requirements for crew sleep and other tasks, yet the number of light-emitting diode (LED) indicators and displays within a habitable volume is currently uncontrolled. Because each of these light sources has its own unique spectral properties, the additive lighting environment ends up becoming something different from what was planned or researched. Restricting the use of displays and indicators is not a solution because these systems provide beneficial feedback to the crew.
The research team for this grant used computer-based computational modeling and real-world lighting mockups to document the impact that light sources other than the ambient lighting system contribute to the ambient spectral lighting environment. In particular, the team was focused on understanding the impacts of long-term tasks located in front of avionics or computer displays. The team also wanted to understand options for mitigating the changes to the ambient light spectrum in the interest of maintaining the performance of a lighting countermeasure. The project utilized a variety of physical and computer-based simulations to determine direct relationships between system implementation and light spectrum.

Using real-world data, two computer models were built in the commercially available optics analysis software Zemax Optics Studio©, one of a glove box-sized volume and the other modeling a real-world-sized environment. The team also built a mockup test facility that had the same volume and configuration as our larger Zemax model. The team collected over 1200 spectral irradiance measurements, each representing a different configuration of the mockup. Analysis of the computer model data and real-world data showed a measurable impact on ambient light spectrum. This data showed that obvious design techniques exist that can be used to bind the ambient light spectrum closer to the planned spectral operating environment for the observer’s eye point.

In summary, the following observations should be considered when designing an operational environment that is dominated by computer displays. The more light a light source directs into the field of view of the observer, the more of an impact it will make on various human factors issues that depend on spectral shape and intensity. Because viewing angle has a large part to play in the amount of light flux on the crewmember’s retina, beam shape, combined with light source location is an important factor for determining percent probable incident flux on the observer from any combination of light sources. Display systems are particularly good at collimating light and maintaining image coherency with respect to viewing angle. As a result, computer graphics design and display lumen output are major factors influencing the amount of spectrally intense light projected into the environment and in the viewer’s direction. Use of correlated color temperature (CCT) adjustable white point display software was useful only if the predominant background color was white and if it matched the ambient light system’s CCT. It was discovered that display graphics that used a predominantly black background had the least influence on unplanned spectral energy projected into the environment. Data from computer models and physical testing showed that percent reflectance makes a difference in total energy reflected back into an environment and that reflectance, within certain architectural geometries, can be used to control the amount of a light spectrum that is allowed to perpetuate in the environment. Both physical and modeled data showed that room volume and distance from significant light sources influence the total spectrum in a room. When a high surface reflectance is used, the smaller environments had a homogenizing effect on total light spectrum, whereas larger environments were less mixed.

The findings indicated above should be considered when making recommendations for practice or standards for architectural systems. The ambient lighting system, surface reflectance, and display and indicator implementation all factor into the users’ spectral environment. A variety of low-cost solutions exist to mitigate the impact of light from non-architectural lighting systems, and much potential for system automation and integration of display systems with the ambient environment. This team believes that proper planning can be used to avoid integration problems and also believes that human-in-the-loop evaluations, real-world test and measurement, and computer modeling can be used to determine how changes to a process, display graphics, and architecture will help maintain the planned spectral operating lighting environment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research we performed on this grant, both the computer and real world simulations, showed that light sources not part of the ambient lighting system do contribute and modify the final light spectrum of the environment. The study also showed that there are ways to control these impacts, pending on the surface materials, the operational volume, the astronaut's viewing angle of non architectural lighting systems, the spectrum of the ambient lighting system and displays, and the graphics chosen for computer displays.
The challenge, however, is that there are many variables at play. Recommendations for any element of the environment or crew task at a high level may be difficult. The biggest finding is that the power of the light source, the direction of the flux, distance from a light source, and diffusive properties of the environment are important factors and may be able to have some sort of risk factors assigned.

Any improvements recommended by the research from this project can be immediately applied to architecture design techniques for Earth based applications.

Application of Findings to Future Requirements and Design Implementations
With more time and funding, much more study of computer models and real-life architectural settings could have been performed on the data collected. Sufficient data was obtained from both efforts, however, to show trends and to elucidate design practices to avoid and architectural innovations to pursue in a microgravity environment.

Light Source Power

The more light a light source directs into the field of view of the observer, the more of an impact it will make on various behavioral health and human factors issues that depend on light spectrum and intensity. Assuming an environment with a predominantly white reflectance spectrum, a simple initial evaluation would be to compare the total spectral power per wavelength of each planned light source. Spectral powers can be summed to show a predicted integrated spectral power. If multiple significantly different light source spectrums exist within the environment in addition to the ambient lighting system, the total spectral distribution will be significantly changed.

Task Progress:

Light Source Beam Shape

Because viewing angle has a large part to play in the amount of light flux on the retina, beam shape combined with light source location is an important factor for determining percent probable incident flux on the observer from any combination of light sources. If a particular lamp, display, or LED generates an offending light spectrum, consideration of the total light power output of the source plus creative placement of these light sources with respect to the ambient lighting system is a countermeasure. For example, LEDs used as colored indicator lamps should be selected to emit the minimal amount of light needed for the task required. Additionally, task analysis should be done to select LED indicator beam angle and intensity for the tasks required. This will not only reduce the amount of unwanted light, but potentially reduce power usage as well. To reduce the percentage of additional retinal exposure from an LED indicator, if an LED indicator is within the direct field of view, its output illuminance at the observer's eye point (vertical illuminance or irradiance) should be much lower than the vertical illuminance contributed by the ambient lighting system as the same observation point.

Collimation of Light

Display systems are particularly good at collimating light. The optical systems on computer display monitors are designed to provide a wide viewing angle while maintaining optical coherency of the image projected on the display. Depending on the method of optical design, the optics may have diffractive properties that cause spectral shift depending on viewing angle. The designer needs to expect that the spectrum measured from any grouping of pixels on the display will project into the room at any viewing angle within the display's designed viewing angle specification. Because of these features, the location of collimating light sources, like displays, within the viewer's operational environment will have an impact on how the rest of the environment factors into the total spectrum the viewer sees.

Collimation of light from display systems can also create problems for tasks that require interpretation of a white environment that was designed to represent a specific color temperature. Smaller environments that utilize a neutral reflective color, like white, will homogenize the saturated colored light from a display. Homogenization of colored light with a known white light source may influence human and vision system "white balance" interpretations. The field of view of the viewer is a potential countermeasure. If the viewer observes a test article primarily lit by the ambient lighting system, and the ambient lighting system is significantly brighter than the colored display, the color impact from the display should be less. Training crewmembers to be observant of the lighting systems within a task area and providing the option of having an ambient light source at all major or critical surfaces are possible countermeasures.

Display Graphics

Both physical and computer-simulated data showed that display graphics design had a significant impact on the spectral power output from a display. The more pixels used on the display, especially pixels that combined more than one of the red-green-blue primaries, the more energy was emitted. White screens emitted the most light. It was also found, however, that for applications that needed to use a predominantly white screen, use of CCT-adjustable white point software applications was able to significantly reduce melanopic lux. However, if the intent is to design an avionics or telerobotics console with displays that the user is in front of for a significant amount of time, the background should be predominantly black, and if white is used, it should attempt to match the CCT of the ambient lighting system. If the displays are intended to be used before the user attempts to sleep, displays should be designed with predominantly black screens so that the ambient lighting environment is the primary sleep and circadian lighting countermeasure.

Surface Reflectance

Data from computer models and physical testing shows that percent surface reflectance of a material makes a difference in total energy reflected back into an environment, especially when the offending light source is directly opposite a reflective panel with low reflectance values. When light from an offending light source hits a dark panel, only a minimal amount is reflected into the space to reflect off the opposite wall and into the eyes of the observer. While placing dark surfaces around a light source will not make an impact on light reflected back to the observer, it does help with visual contrast, which can allow the designer to select indicators and displays with lower output light flux.

Room Volume

Both physical and modeled data showed that room volume and distance from significant light sources influence the total spectrum. Smaller volumes create an integration effect caused by the proximity of reflective surfaces that redirect light from various sources within the environment. With larger volumes, proximity to surfaces is an important factor only for tasks where the viewer is near and facing a large architectural surface. Additionally, with larger volumes the field of view of the observer with respect to major light sources is a larger factor than the reflective surfaces around them. With larger volumes, the user has a higher probability of being behind an overhead ambient light source than in front, allowing flux from the lamp to directly impact the eyes.

Solutions Optimized for Circadian Entrainment

As can be observed from the findings in this report, all elements of the architectural system design factor into the success of a circadian health countermeasure. Display design, graphics content, device and indicator location, surface materials, and ambient lighting system lamp placement all factor into the final retinal light exposure to any individual within a habitable environment. Efforts, like the Solid State Lighting Assembly project, provide a great initial implementation of flexible architectures. To complete this flexibility and automation, however, the rest of the system should be integrated. Requirements, and design iterations should be evaluated as an integrated system to determine impacts. Innovations in vehicle avionics and environmental control automation could further help with the fine tuning of beneficial lighting environments. Finally, accurate computer models, mockups, and human-in-the-loop testing completes the verification that a vehicle architecture designed to promote crew circadian health was effective.

Conclusion & Recommended Future Work

At the start of this project, the team wanted to determine how displays and indicators impact a planned ambient light spectrum and if those impacts could be mitigated. The project used both computer models and physical real-world testing to identify impacts and potential countermeasures. The team determined that the energy and spectral output from the display or indicator, along with proximity to a crewmember's field of view, was directly related to the increase in unwanted spectral impacts.

	Our study collected a vast amount of data that can be mined in the future to look for more relationships between volume, surface reflectance, field of view, and display design. Even though our team collected a large amount of data for a variety of display configurations and ambient room lighting configurations, it was collected from only three viewing angles and two major reflectance configurations of the mockup test facility. Additional studies could be performed, with the data set collected during the course of this project, to investigate how lamp locations, room shape, background luminance, and light source energy output factor into the retinal light exposure of crewmembers. Additional room architecture evaluations should be considered, since computer models did indicate room volume impacts the integration of total light within an environment. Our team recommends that additional resources be allocated for future architectural mockup and computer model simulations for larger and more complex operational environments. Computer modeling of a variety of interior environment models may be more cost effective means of exploring architecture design impacts than real world mockup evaluation alone. Interesting automation research can also be performed to develop a closed-loop dynamic lighting environment where displays and ambient light systems are programmatically synchronized and utilize closed-loop sensor feedback. Finally, finding that the display design had a significant impact on the quantity of additional light transmitted to the environment, this team recommends a future human-in-the-loop study to quantify sleep, circadian, neuroendocrine, visual, and neurobehavioral responses to typical spacecraft avionic and robotics workstation-centric operating environments.
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